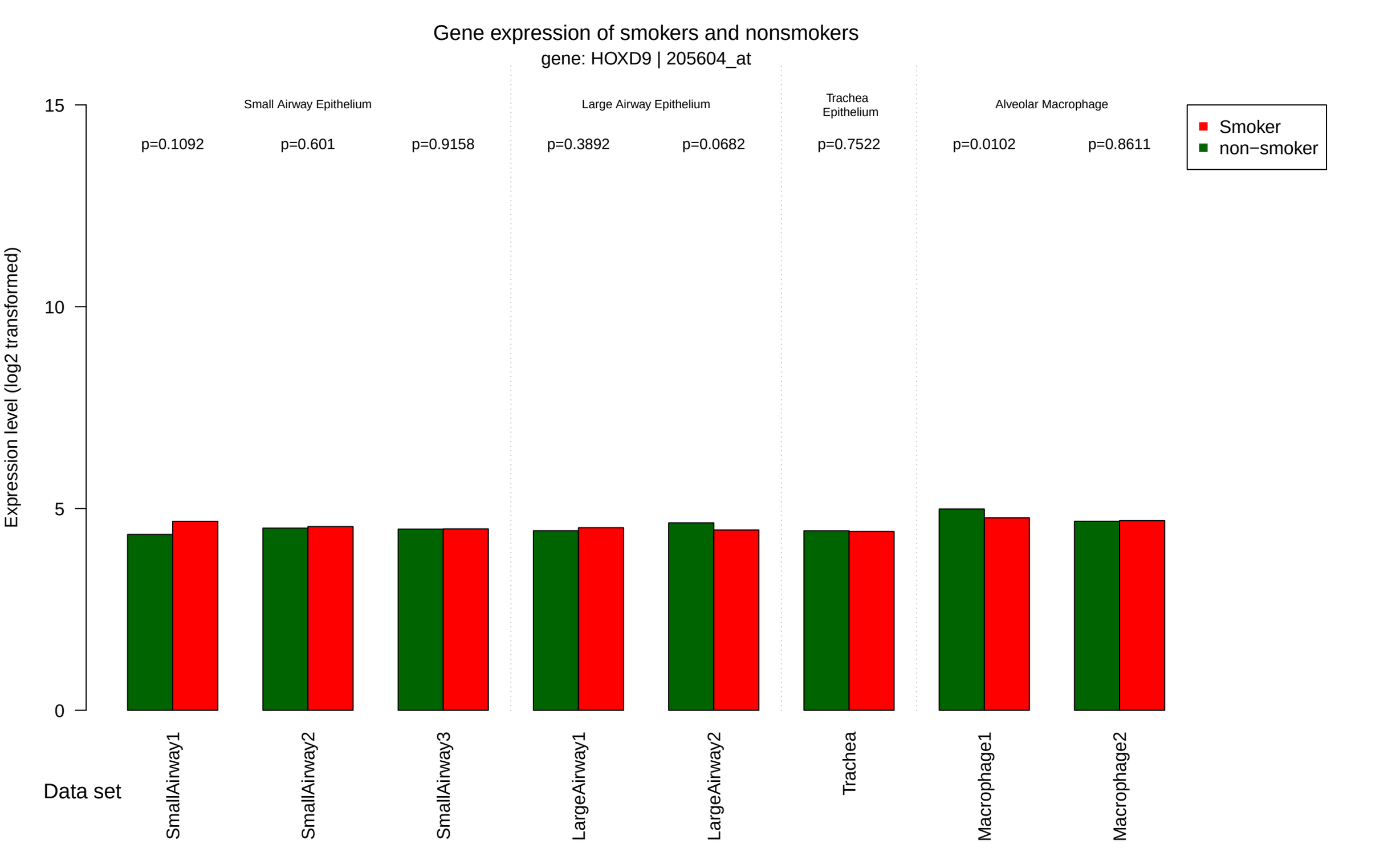
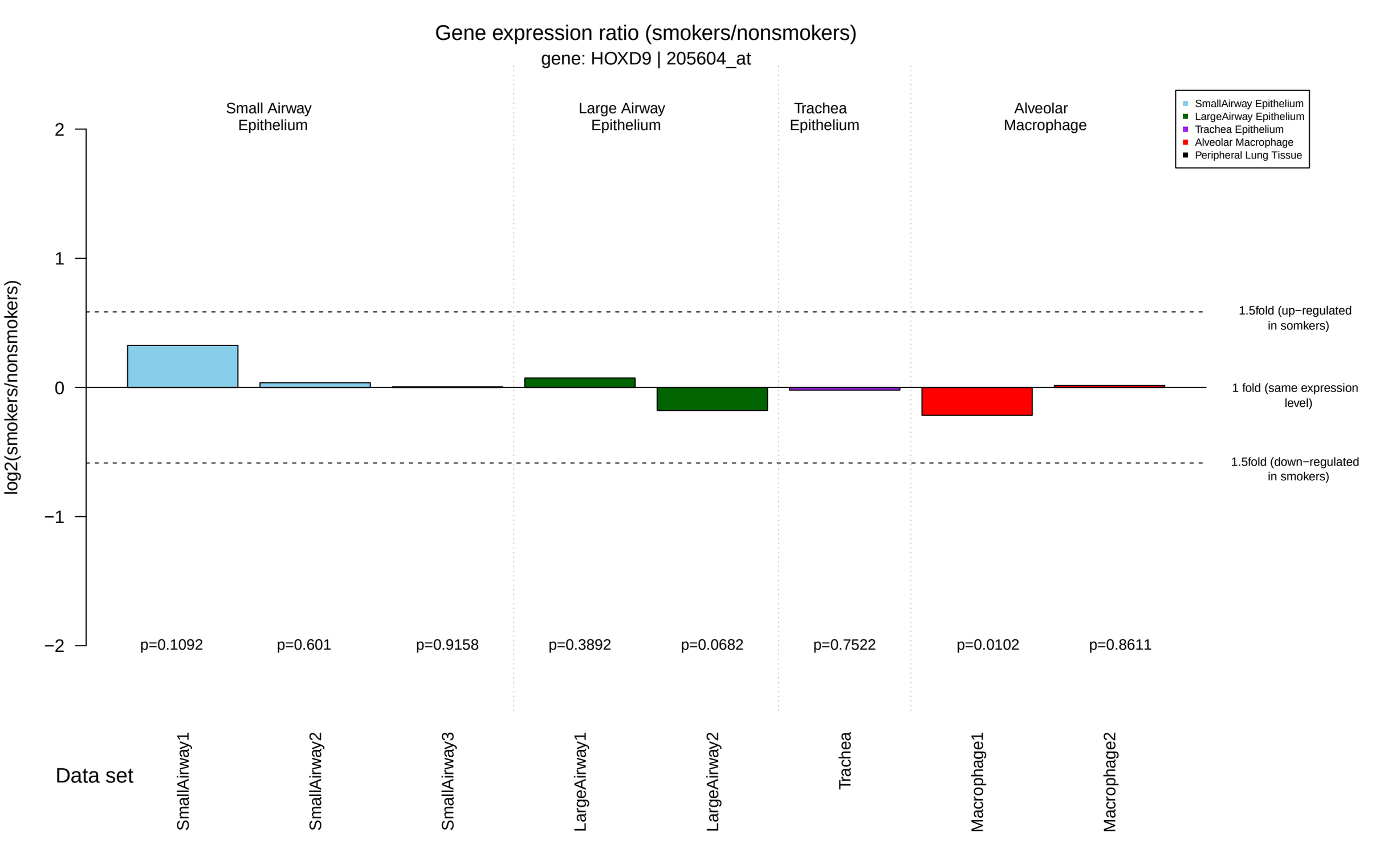
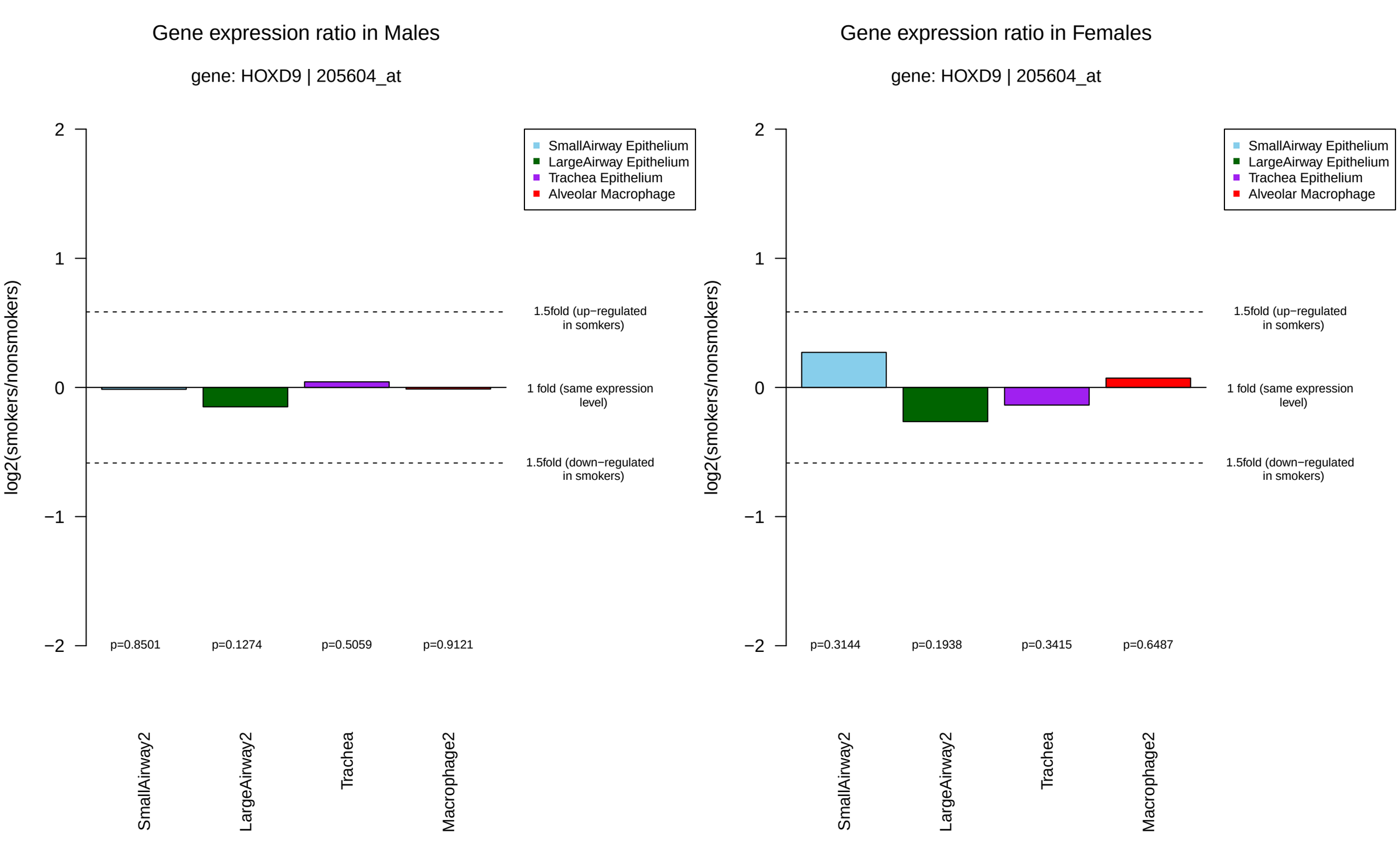
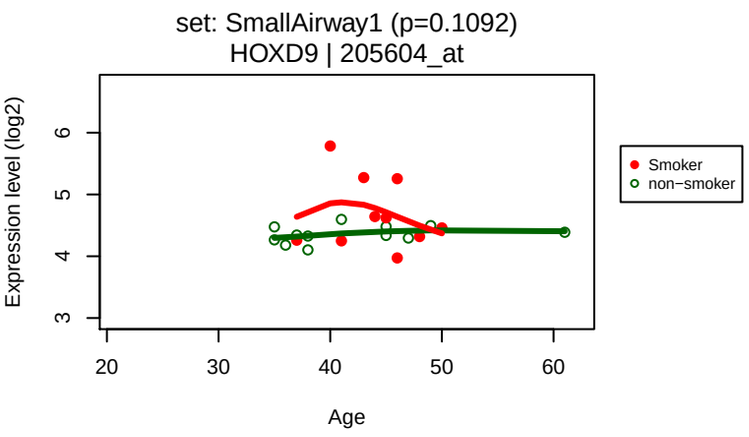
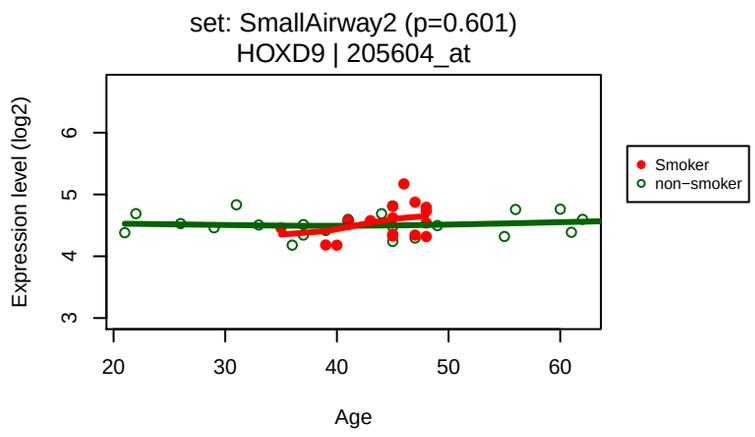
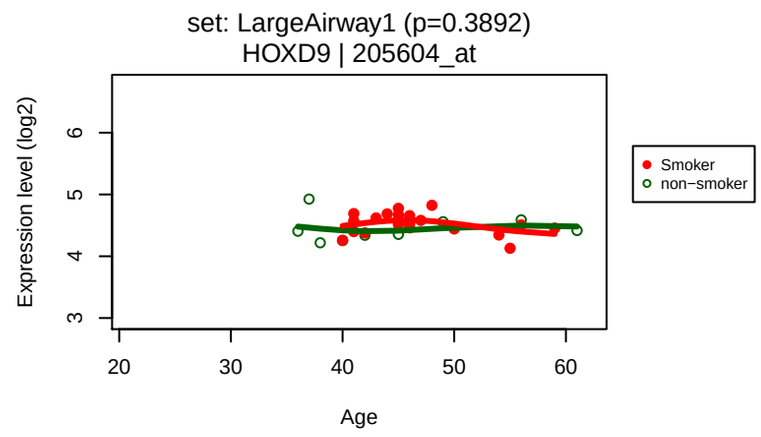

Gene expression of smokers and nonsmokers
gene: HOXD9 | 205604_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1092
p=0.601
p=0.9158
p=0.3892
p=0.0682
p=0.7522
p=0.0102
p=0.8611
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

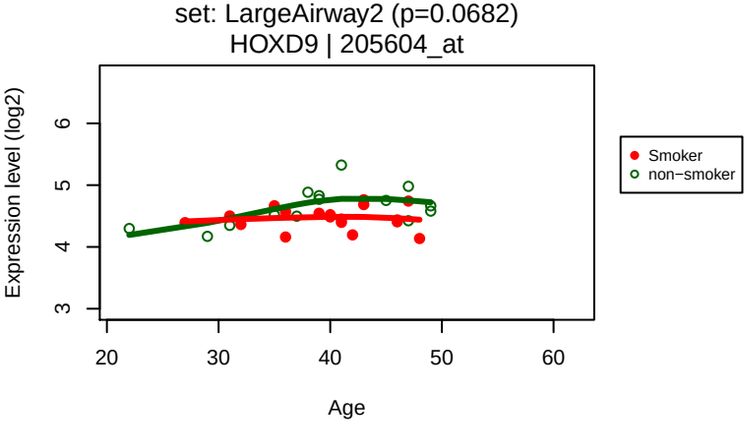

Gene expression ratio (smokers/nonsmokers)
gene: HOXD9 | 205604_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1092
p=0.601
p=0.9158
p=0.3892
p=0.0682
p=0.7522
p=0.0102
p=0.8611
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

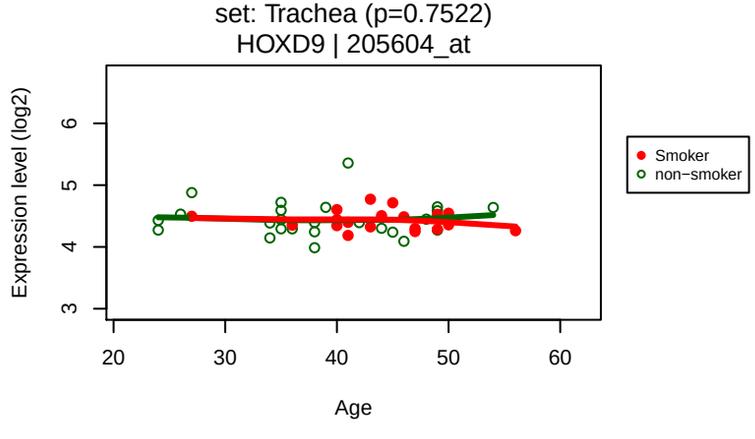

Gene expression ratio in Males
gene: HOXD9 | 205604_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8501
p=0.1274
p=0.5059
p=0.9121
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD9 | 205604_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3144
p=0.1938
p=0.3415
p=0.6487
SmallAirway2
LargeAirway2
Trachea
Macrophage2

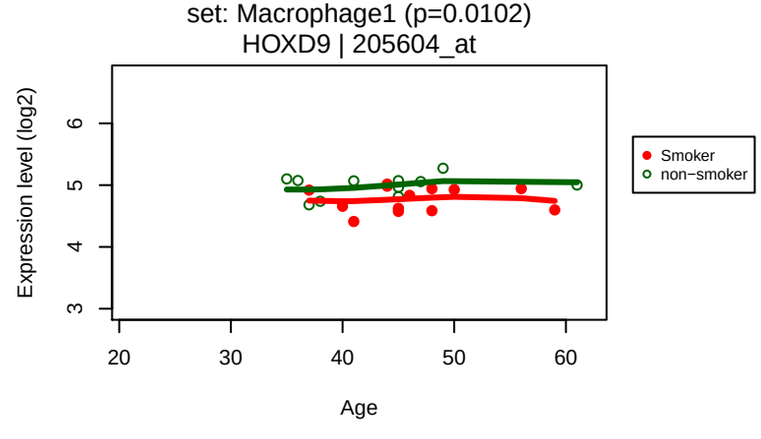

set: SmallAirway1 (p=0.1092)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker

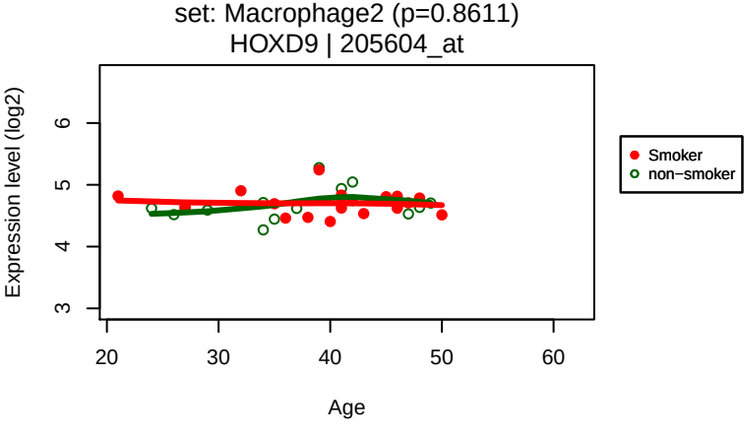

set: SmallAirway2 (p=0.601)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker

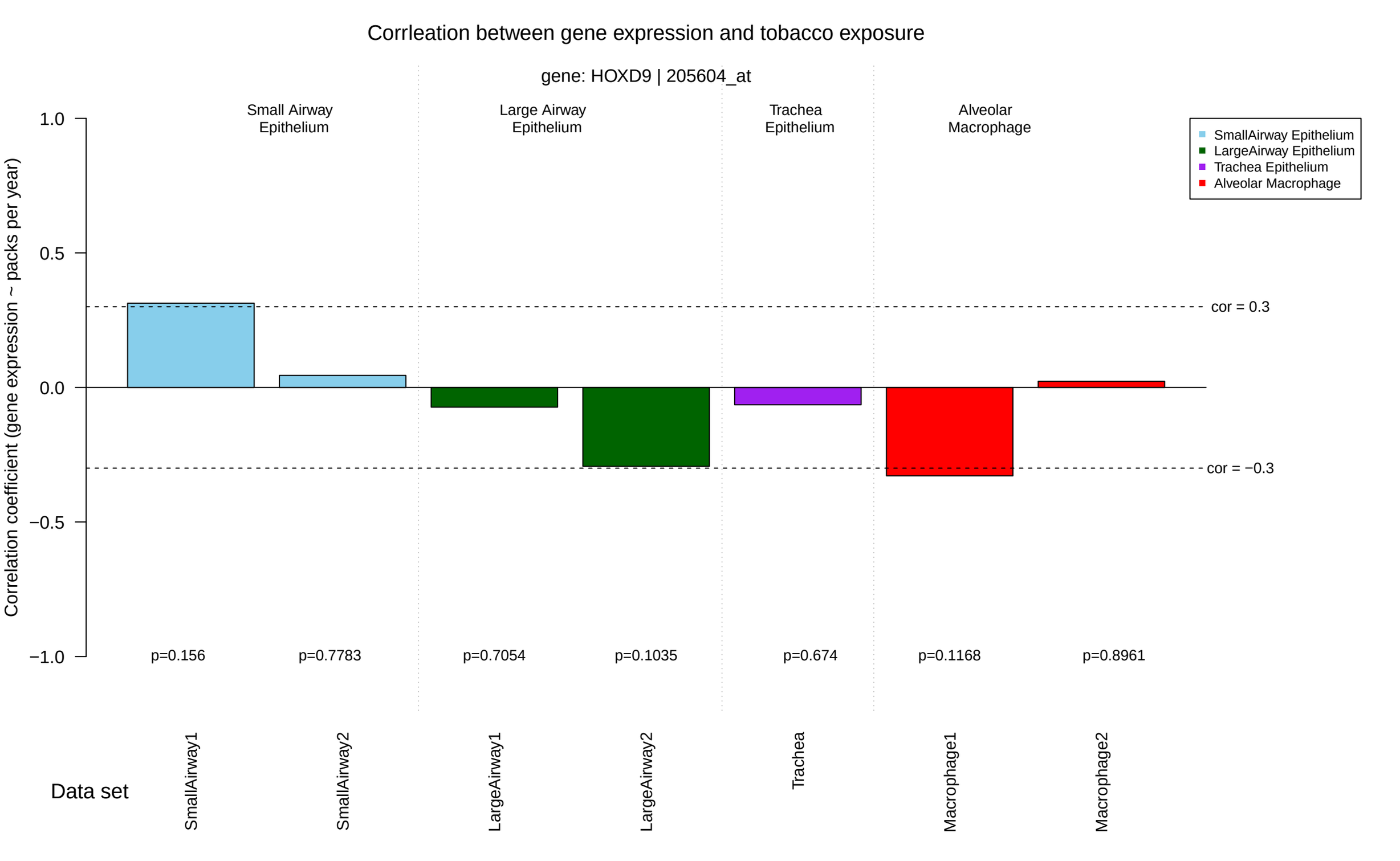

set: LargeAirway1 (p=0.3892)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0682)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7522)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0102)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8611)
HOXD9 | 205604_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD9 | 205604_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.156
p=0.7783
p=0.7054
p=0.1035
p=0.674
p=0.1168
p=0.8961
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2